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Comparison of electron temperature and density measured by helium line intensity ratio and Thomson scattering methods in ECH spherical tokamak plasma

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Abstract

The electron temperature and density profiles in the midplane of a spherical tokamak plasma produced by electron cyclotron heating (ECH) in Q-shu University experiment with steady-state spherical tokamak (QUEST) are measured by the helium line intensity ratio method. The measured profiles are compared with those obtained by the Thomson scattering method, and the measured temperatures and densities are found to agree within factors of ∼2 and ∼6, respectively. Taken together with the previous results of comparisons performed in the scrape-off layers of several toroidal devices, the same degree of agreement between the helium line intensity ratio method and other methods is obtained in the ranges of 7–100 eV for temperature and 4×10^{16}–1×10^{19} m^{-3} for density.

Keywords: spectroscopy, helium line intensity ratio, SOL, QUEST, ECH, spherical tokamak

1. Introduction

In tokamak fusion DEMO reactors under conceptual design [1], the erosion of diagnostic first mirrors by impingements of high-energy charge exchange atoms [2] is believed to become a significant issue accompanying the increase in discharge duration [3, 4]. As a countermeasure to this issue, installing the first mirrors behind a gas-filled long duct having $L/\rho = 40$, 50, and 80, where L and ρ are the length and radius of the duct, respectively, is proposed for infrared, visible, and UV light measurements, respectively, in an EU DEMO reactor [3]. The duct installation, however, greatly reduces the solid angle of light measurements [4] and may make Thomson scattering measurements, for which the use of a large mirror is essential, particularly difficult or may limit measurement

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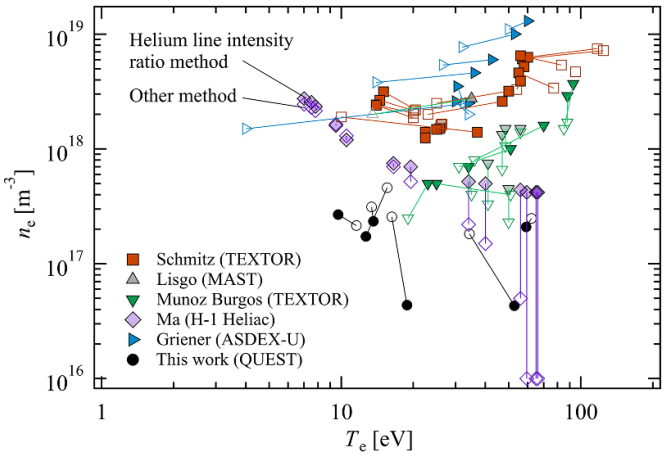

Figure 1. T_e and n_e measured by the helium line intensity ratio (filled symbols) and other (open symbols) methods [9–14]: Thomson scattering, Langmuir probe, microwave interferometry, and lithium beam [9–14]. The data points measured at the same location are connected with a line. For the part of the data points reported by Munoz Burgos [12] (five pale green symbols) and all data points reported by Ma [13], only n_e is measured by the other methods.

locations. Several measures to overcome the limitation of the Thomson scattering method are under consideration, for instance, the use of a line-integrated measurement to increase the amount of light collection [5] and alternative diagnostics such as microwave interferometry/reflectometry [6], infrared laser polarimetry [7], and helium line intensity ratio method [8].

With these as background, in this study, we aim to obtain experimental data on the performance of the helium line intensity ratio method in T_e and n_e measurements in a toroidal plasma, where T_e and n_e are the electron temperature and density, respectively. The technique can be used to evaluate the scrape-off layer (SOL) and divertor plasmas [9–17], and implemented only by passive emission spectroscopy, which is compatible with light collection through the ducts. Thus far, the method has been applied to various plasmas, and among them, recent results of comparisons with other methods [9–14] are plotted in figure 1. The figure summarizes comparisons carried out using toroidal plasmas and spatially resolved values. The comparisons were made using various methods, such as the Thomson scattering method, the Langmuir probe method, microwave interferometry, and the lithium beam method, the T_e and n_e values obtained by the helium line intensity ratio method agree with those obtained by the other methods within a factor of ∼8 for both T_e and n_e. Note that some data with a particularly large n_e difference obtained using H-1 heliac were excluded.

T_e and n_e evaluated in the previous studies ranged from 7 eV to 100 eV and 4×10^{17} m^{-3} to 1×10^{19} m^{-3}, respectively. In this study, we additionally provide comparison data performed in a smaller n_e condition from 4×10^{16} m^{-3} to 2×10^{17} m^{-3}, which would be useful for the measurements of plasmas for discharge wall cleaning and discharge initiation. We conducted an experiment using a spherical tokamak plasma produced by electron cyclotron heating (ECH) in Q-shu University experiment with steady-state spherical tokamak (QUEST) [18], and performed Thomson scattering measurements for the comparison with the helium line intensity ratio method.

2. Experiment

Schematics of the QUEST poloidal section and midplane are shown in figure 2. A spherical tokamak plasma was produced by ECH at 8.2 GHz and 22 kW for 65 s. In the midplane, the ECH resonance layer was at $R = 0.28$ m (red dashed line) near the center stack (CS) surface at $R = 0.22$ m, where R is the major radius. The plasma's last-closed flux surface (LCFS) was limited by the CS, and the positions of the magnetic axis and outer LCFS were approximately 0.55 m and 0.91 m, respectively. The radiation shield (RS) was at $R = 1.3$ m.

To reproduce plasmas containing a small amount of helium, helium was first adsorbed on the plasma-facing components (PFCs) by producing helium-puffed hydrogen plasmas. Then, hydrogen plasmas without helium puffing were produced, and emission from helium atoms desorbed from the PFCs was measured. Three consecutive discharges (#48921–48923) after the helium adsorption procedure were used for the measurements. The discharges became steady after ∼20 s, providing good reproducibility of plasma parameters and helium atom spectra. T_e and n_e profiles in the midplane measured by the Thomson scattering method [19, 20] at 62.5 s are indicated by symbols in figure 3. T_e reached up to 60 eV on the inboard side near the ECH resonance layer, and n_e had a relatively flat profile of 2–5 $\times 10^{17}$ m^{-3}. These profiles are general characteristics of the QUEST 8.2 GHz ECH plasma [21]. For the Thomson scattering measurement, a backward scattering configuration, in which the scattering length and scattering intensity increase, is adopted to measure low-density plasmas [19]. Furthermore, the measurement in this study was performed at every 1 s with accumulating ten spectra measured at every 100 ms interval to improve the signal-to-noise ratio of the spectra. The temporal stability of the measurement was previously confirmed in a long-duration discharge plasma, having T_e and n_e similar to the present plasma [20]. The hydrogen and helium atom densities evaluated using a quadrupole mass spectrometer at the vacuum chamber [22] were 7×10^{16} m^{-3} and 5×10^{14} m^{-3}, respectively, whereas the helium atom density evaluated by spectroscopy in this study was 2–5 $\times 10^{14}$ m^{-3}.

Spatially resolved spectroscopy was conducted using fan-shaped 22 line-of-sights (LOSs) in the midplane. Two sets of LOSs were used, each composed of an achromatic lens (15 mm focal length) and a bundle of 15 optical fibers (230/250 μm core/cladding diameters, 0.2 NA). Among the 30 fibers in these two sets, 22 were used for this study. The LOSs were

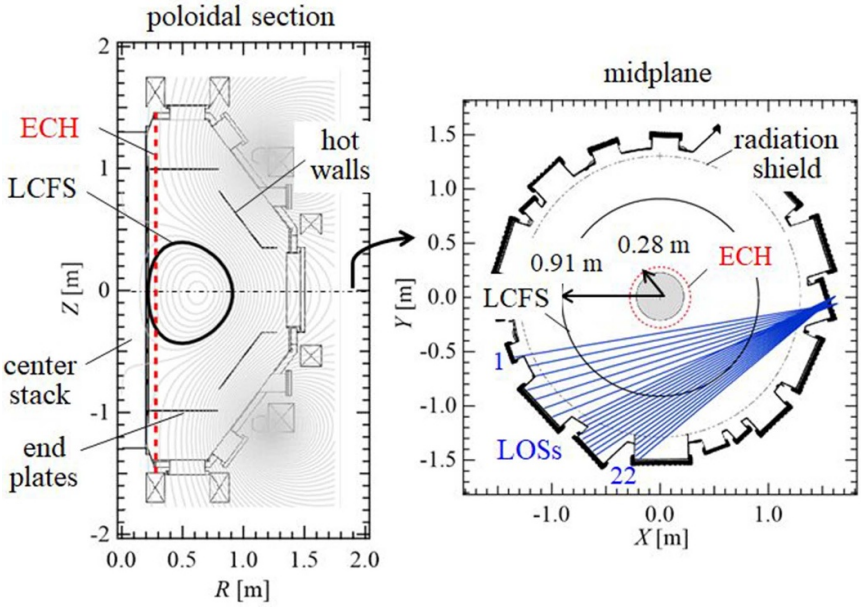

Figure 2. Poloidal section (left) and midplane (right) of QUEST. The ECH resonance layer is shown with the red dashed line, the last-closed flux surface (LCFS) with the black solid line, and magnetic flux surfaces with gray solid lines. The 22 line-of-sights (LOSs) shown with blue solid lines were used for spectroscopy. The LOSs were numbered 1–22 from the inboard side.

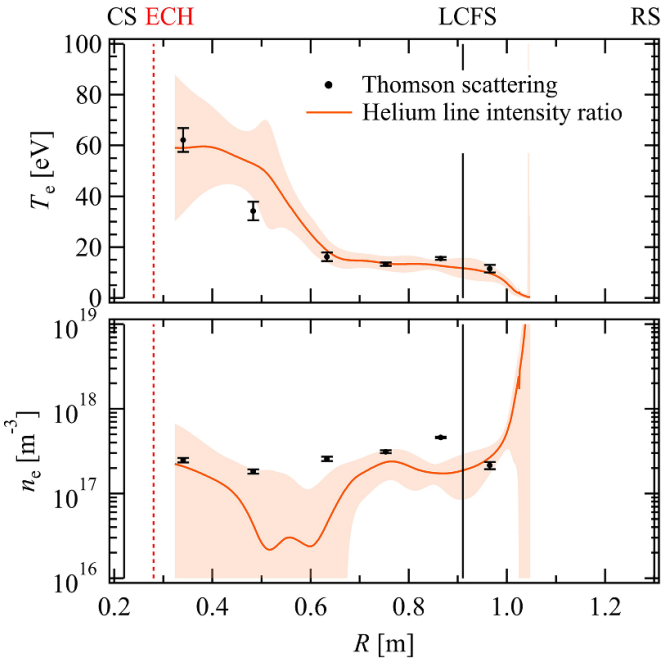

Figure 3. T_e and n_e profiles in the midplane measured by the Thomson scattering (circles) and helium line intensity ratio (orange solid lines) methods. The shaded region indicates the range of the error bar; this plotting method is also used in the other figures. The positions of the CS surface, ECH resonance layer, outer LCFS, and RS are indicated by vertical lines.

numbered 1–22 from the inboard side, and the distance from the tokamak Z-axis was $\eta = 0.32$–1.02 m. The diameter of the LOSs was approximately 30 mm at the positions closest to the Z-axis. A custom-made spectrometer (105 mm focal length, 420 grooves per mm grating, F/1.4) combined with a CMOS camera (Basler a2A5320-23umPRO, 5328 × 3040 pixels, 2.7 μm square pixel size, 12 bit) as shown in figure 4 was used to measure spectra for all the LOSs simultaneously. The wavelength range and resolution were 400–730 nm and 0.6 nm, respectively, and the reciprocal linear dispersion was 0.06 nm per pixel at 565 nm. The sensitivity and wavelength of the spectroscopy system were absolutely calibrated. For the former, the transmittance of the window was considered using the Fresnel equations. The effect of reflection on the RS and vacuum vessel surfaces is unknown. Still, we assume the effect is insignificant since significant outliers were not observed in the emission line intensities measured on the 22 LOSs. The spectra measured over 62.0–62.8 s were used for analysis.

3. Results

Figure 5 shows a spectrum measured on LOS1, in which H I, He I, C II, C III, O I, O II, and O III emission lines were observed. For He I, five lines, 501.6 nm (2^1S–3^1P), 587.6 nm (2^3P–3^3D), 667.8 nm (2^1P–3^1D), 706.5 nm (2^3P–3^3S), and 728.1 nm (2^1P–3^1S), were identified. The intensities of the He I lines were low owing to the small helium atom density. Owing to the superposition of C II and O II emission lines, the intensities of only three He I lines, 587.6 nm, 667.8 nm, and 706.5 nm, could be accurately evaluated. Note that for the 706.5 nm line, the intensity of a superposed C II line (3p $^4S_{3/2}$–3d $^4P^\circ_{5/2}$) was evaluated from those of the other C II lines having the identical upper electronic state assuming the statistical distribution of the upper J-state population densities, where J is the total angular momentum quantum number.

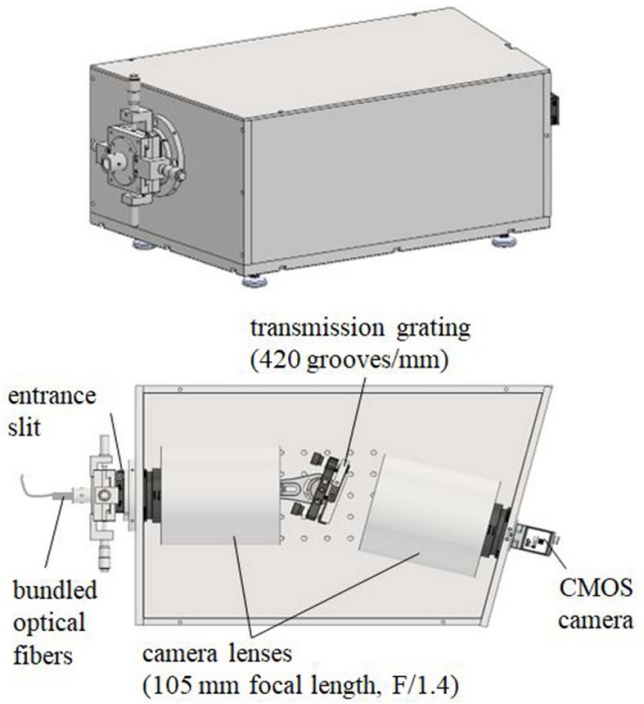

Figure 4. Schematic of the spectrometer (overview of the top and inside view of the bottom), which consists of two camera lenses, transmission grating, and a CMOS camera. Spectra for all the LOSs can be measured simultaneously.

The radiance (line-integrated intensity) of the three He I lines was evaluated from the area of the line spectra and is plotted with the symbols at the top of figure 6. The abscissa of the figure is η. The symbols and error bars are the means and standard deviations of three consecutive discharges, respectively. The latter was used to consider the reproducibility of the spectroscopy. The emissivity (local intensity) was then evaluated from the radiance using the Abel inversion, where the axial symmetry of the plasma was assumed, and the effect of the LOS finite diameter was neglected. Note that the Abel inversion is simple but requires axisymmetry of the emission intensity, and this limits the application of the method only on the midplane. For measuring the other part of the plasma, the tomographic reconstruction method can be an alternative, *e.g.* [17, 23]. The following procedure was applied to a given He I line to evaluate the mean and standard deviation of emissivity [24].

(1) We denote the radiance and its standard deviation measured on an LOS at the distance η as $I_{\mathrm{exp}}(\eta)$ and $\sigma_{\mathrm{exp}}(\eta)$, respectively. The 22 data points of $I_{\mathrm{exp}}(\eta)$ are interpolated by a smooth spline function with the smoothing parameter of 1.5, and the outer boundary of the emission η_{edge} is determined as the position where the tangent of the spline function at the outermost data point ($\eta = 1.02$ m) intersects with zero radiance level.
(2) At the 22 η-positions of $I_{\mathrm{exp}}(\eta)$, 10^3 pairs of artificial radiance data are randomly generated such that the mean and standard deviation of the data coincide with $I_{\mathrm{exp}}(\eta)$ and $\sigma_{\mathrm{exp}}(\eta)$, respectively. Each artificial radiance data is interpolated by a smooth spline function in the range $\eta = 0.32$ – η_{edge} to satisfy the boundary condition of zero radiance at η_{edge}. This procedure produces 10^3 continuous curves of radiance $I_{\mathrm{ss}}(\eta)$.
(3) The Abel inversion expressed by the following equation is applied to $I_{\mathrm{ss}}(\eta)$.

$$\epsilon_{\mathrm{ss}}(R) = -\frac{1}{\pi}\int_{R}^{\eta_{\mathrm{edge}}} \frac{\mathrm{d}I_{\mathrm{ss}}(\eta)}{\mathrm{d}\eta}\frac{\mathrm{d}\eta}{\sqrt{\eta^2 - R^2}}. \quad (1)$$

This procedure produces 10^3 continuous curves of $\varepsilon_{\mathrm{ss}}(R)$. The mean and standard deviation of the emissivity are calculated from the $\varepsilon_{\mathrm{ss}}(R)$ curves.

The obtained emissivity of the three He I lines is shown at the bottom of figure 6. The emissivity peaked near the outer LCFS, decreased in the core owing to a decrease in the helium atom density caused by ionization, and increased on the inboard side. This tendency is consistent with a physical image that helium atoms are desorbed from the PFCs and flow into the plasma while being ionized. Note that H_δ emissivity measured in the 28 GHz ECH plasma has a similar profile [25].

4. Discussion

From the ratios of the measured emissivity shown at the bottom of figure 6, T_{e} and n_{e} profiles were evaluated. The ratio of the spin-singlet to spin-triplet He I emission lines, 667.8 nm/587.6 nm, is sensitive to T_{e} because of the different T_{e} dependences of the electron-impact excitation rate coefficient from the ground state. On the other hand, the ratio of the same spin lines, 587.6 nm/706.5 nm, is sensitive to n_{e} because the relative population densities of high energy levels increase with n_{e}. The emissivity ratios calculated using a collisional-radiative (CR) model [26] are shown in figure 7, where the detailed calculation conditions are described in the appendix. When $T_{\mathrm{e}} > 10$ eV and $n_{\mathrm{e}} > 8 \times 10^{16}$ m^{-3}, the ratios 667.8 nm/587.6 nm and 587.6 nm/706.5 nm depend nearly only on T_{e} and n_{e}, respectively, and T_{e} and n_{e} can be determined from the emissivity ratios, and vice versa. On the other hand, when $T_{\mathrm{e}} < 10$ eV and $n_{\mathrm{e}} < 8 \times 10^{16}$ m^{-3}, the n_{e} dependence is lost since the excitation regime approaches the corona equilibrium [27], where the emissivity is proportional to n_{e}, and only T_{e} can be determined.

T_{e} and n_{e} at each R-position were evaluated by the least-squares fitting of the emissivity ratios calculated using the CR model to the measured ones. The measured emissivity ratios and the fitting results at the R-positions of the Thomson scattering measurements are shown in figure 8, where the residual of the fitting is small. The obtained T_{e} and n_{e} profiles are shown in figure 3. Their errors mainly originate from the errors of the measured emissivity ratios and the sensitivity of the emissivity ratios to the variations in T_{e} and n_{e}. The obtained T_{e} and n_{e} are close to those measured by the Thomson scattering method. For the results obtained by the helium line intensity ratio method, an increase in n_{e} in the outer edge region

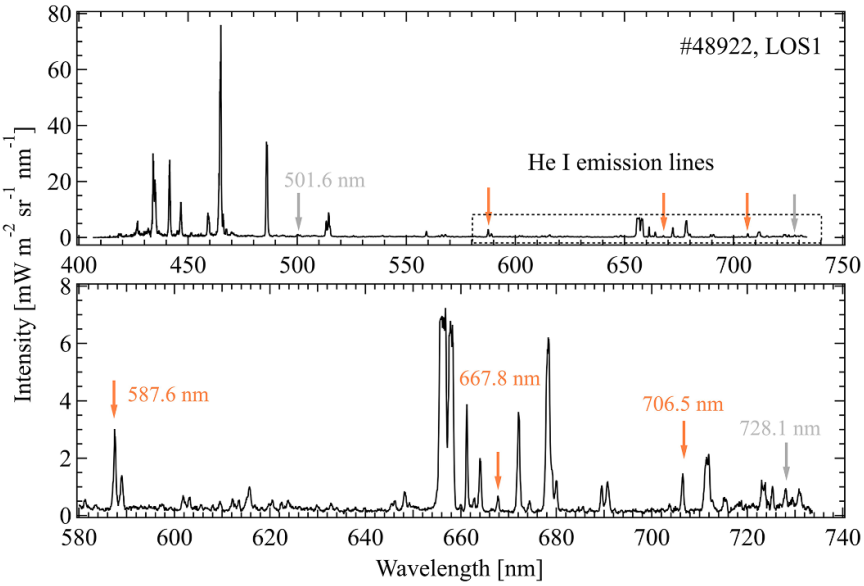

Figure 5. Spectrum measured on LOS1 (top). The figure at the bottom is an enlarged spectrum in the range indicated by the dashed line in the top figure. The five He I emission lines denoted by arrows were identified, and the three He I emission lines indicated by orange arrows were used for analysis. The other HeI emission lines were attributed to H I, C II, C III, O I, O II, and O III. The line spectra of large intensities (H_α, H_β, and some C II and C III lines) were intentionally saturated to obtain sufficient CMOS counts for the He I lines. The saturation level varied with the wavelength owing to intensity calibration.

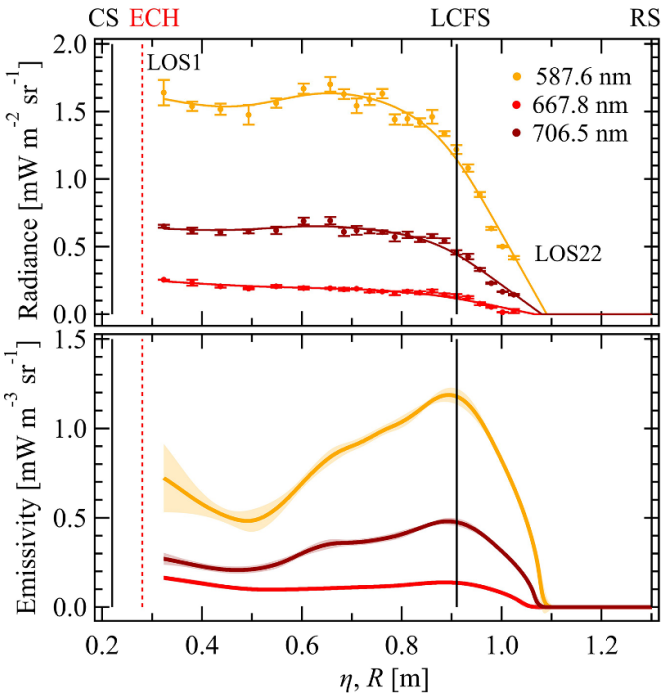

Figure 6. Radiance (line-integrated intensity) (top) and emissivity (local intensity) (bottom) of the three He I emission lines. The abscissa of the top figure is the distance η of the LOS from the Z-axis, and that of the bottom figure is the major radius R.

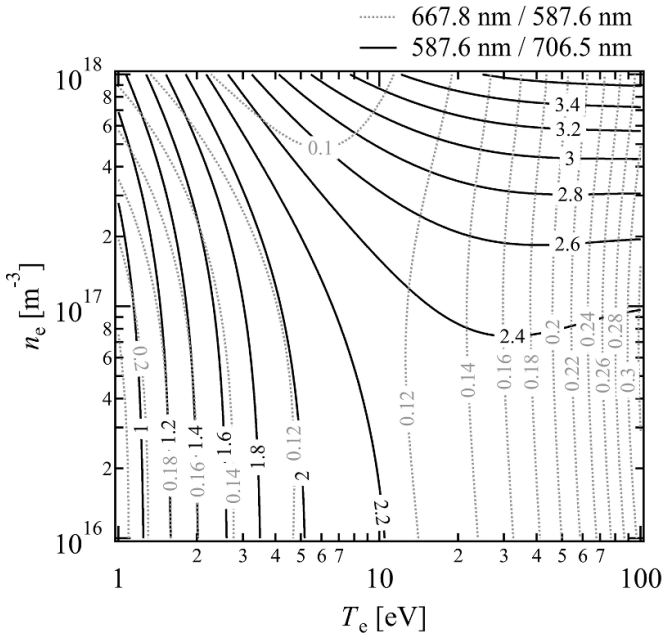

Figure 7. Emissivity ratios of 667.8 nm/587.6 nm and 587.6 nm/706.5 nm for He I emission lines as functions of T_e and n_e calculated using a CR model.

($R > 1.0$ m) is physically implausible. In this region, the excitation regime might have approached the corona equilibrium. Moreover, the larger error bars of both T_e and n_e on the inboard side originate from the error bars of the emissivity. The ground state helium atom density evaluated from the emissivity using the CR model calculation was 2–5 $\times$ 10^{14} m^{-3} at the R-positions of the Thomson scattering measurements.

The mean values of T_e and n_e measured by the Thomson scattering and helium line intensity ratio methods are plotted in figure 1. The agreement between the two methods was within factors of $\sim$2 for T_e and $\sim$6 for n_e. To quantify the degree of

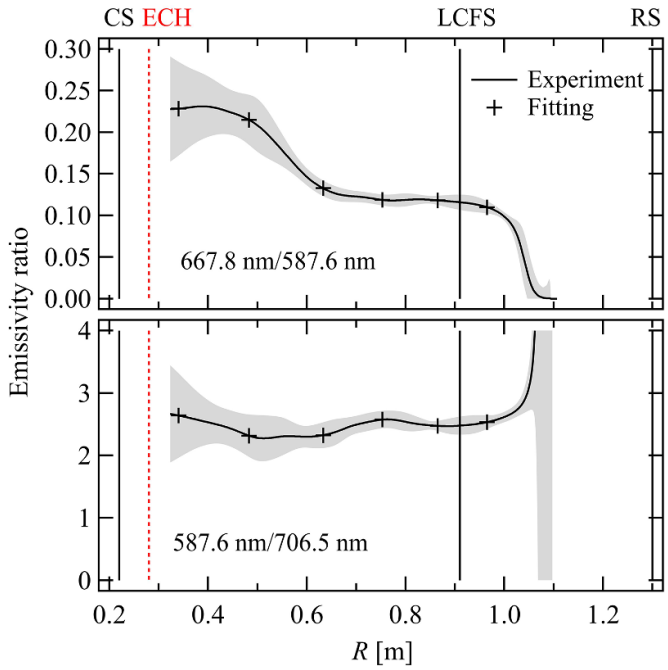

Figure 8. Measured emissivity ratios and the CR model fitting results at the R-positions of the Thomson scattering measurements.

Table 1. Reduced chi-squares of normalized T_e and n_e for the data points plotted in figure 1.

	Schmitz (TEXTOR)	Lisgo (MAST)	Munoz Burgos (TEXTOR)	Ma (H-1 Heliac)	Griener (ASDEX-U)	This work (QUEST)
$\bar{\chi}^2_{T_e}$	6.2×10^{-1}	2.5	2.3×10^{-1}		6.7	6.3×10^{-2}
$\bar{\chi}^2_{n_e}$	5.4×10^{-2}	1.4×10^{-1}	7.7×10^{-1}	4.9×10^{2}	9.9×10^{-2}	3.0×10^{-1}

agreement, we evaluated the reduced chi-squares of the normalized T_e and n_e using the following equations:

$$\bar{\chi}^2_{T_e} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{T^{\mathrm{He}}_{e,i} - T^{\mathrm{other}}_{e,i}}{T^{\mathrm{other}}_{e,i}}\right)^2 \quad (2)$$

$$\bar{\chi}^2_{n_e} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{n^{\mathrm{He}}_{e,i} - n^{\mathrm{other}}_{e,i}}{n^{\mathrm{other}}_{e,i}}\right)^2, \quad (3)$$

where N is the number of data points, $T^{\mathrm{He}}_{e,i}$ and $T^{\mathrm{other}}_{e,i}$ are the electron temperature measured by the helium line intensity ratio and other methods, respectively, and $n^{\mathrm{He}}_{e,i}$ and $n^{\mathrm{other}}_{e,i}$ are the electron density measured by the helium line intensity ratio and other methods, respectively. The values of $\bar{\chi}^2_{T_e}$ and $\bar{\chi}^2_{n_e}$ for the data points plotted in figure 1 are summarized in table 1. The table shows that in this study, T_e measured by the two methods agree better than previous studies, while n_e agree moderately. In addition, a possible tendency observed in this study is that n_e measured by the helium line intensity ratio method are smaller than those measured by the Thomson scattering method. This tendency is contrary to the previous studies. If the assumed statistical distribution of the upper J-state population densities for the C II emission line subtracted from the He I 706.5 nm line is inaccurate, the underestimation of n_e could occur by the emissivity ratio 587.6 nm/706.5 nm. Overall, the same degree of agreement between the helium line intensity ratio method and other methods compared with the previous studies was obtained in a smaller n_e condition from 4×10^{16} m^{-3} to 2×10^{17} m^{-3}.

5. Summary

T_e and n_e profiles in the midplane of a spherical tokamak plasma produced by ECH in QUEST were measured by the helium line intensity ratio method. The measured profiles were compared with those obtained by the Thomson scattering method, and the measured temperatures and densities were found to agree within factors of ~2 and ~6, respectively. Taken together with the previous results of comparisons performed using several toroidal devices, the same degree of agreement between the helium line intensity ratio method and other methods was obtained in the ranges of 7–100 eV for T_e and 4×10^{16}–1×10^{19} m^{-3} for n_e.

It is important to conduct further comparative experiments in the high-density condition corresponding to divertor plasma, where $T_e < 10$ eV and $n_e > 10^{19}$ m^{-3}, to access the applicability of the method to fusion DEMO reactors. Thus far, a favorable experimental result has been reported using a divertor simulator plasma and line-integrated spectra in the n_e range of 10^{19}–10^{21} m^{-3} [28].

Data availability statement

The data cannot be made publicly available upon publication due to legal restrictions preventing unrestricted public distribution. The data that support the findings of this study are available upon reasonable request from the authors.

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Conflict of interest

The authors have no conflicts of interest to disclose.

Appendix. Conditions of calculation using the CR model

In the calculation using the CR model, the effects of radiation trapping, *i.e.* the reabsorption of the photons produced by resonance $1^1\mathrm{S}$–$n^1\mathrm{P}$ ($n \geqslant 2$) emission lines [29–31] and the transport of metastable atoms ($2^1\mathrm{S}$ and $2^3\mathrm{S}$), were neglected. For the former, the optical thickness of an emission line at the center of a uniform cylindrical plasma of radius a and infinite length is expressed as [30]

$$\tau = \frac{e^2}{4m_\mathrm{e}\epsilon_0}\sqrt{\frac{M}{2\pi k_\mathrm{B}T_{1^1\mathrm{S}}}}\frac{f}{\nu}n_{1^1\mathrm{S}}a, \tag{A1}$$

where e is the elementary charge, m_e and M are the electron and helium atom masses, respectively, ε_0 is the vacuum permittivity, and k_B is the Boltzmann constant. In addition, $T_{1^1\mathrm{S}}$ and $n_{1^1\mathrm{S}}$ are respectively the temperature and density of the ground state helium atoms, f is the absorption oscillator strength, and ν is the transition frequency. The substitution of representative values $n_{1^1\mathrm{S}} = 5 \times 10^{14}$ m^{-3}, $T_{1^1\mathrm{S}} = 0.1$ eV, $a = 0.3$ m, $f = 7.346 \times 10^{-2}$, and $\nu = 5.59 \times 10^{15}$ Hz for the $1^1\mathrm{S}$–$3^1\mathrm{P}$ emission line, whose radiation trapping effect is the largest for the present analysis using the CR model, in equation (A1) results in $\tau = 4.0 \times 10^{-4}$ and reabsorption is negligible.

For the latter, the effective electron-impact excitation and ionization rate for the $2^3\mathrm{S}$ metastable atoms is approximately $n_\mathrm{e}\langle\sigma v\rangle_\mathrm{eff} =\sim 10^6$ s^{-1} when $T_\mathrm{e} = 20$ eV and $n_\mathrm{e} = 2 \times 10^{17}$ m^{-3}. If we assume a $2^3\mathrm{S}$ atom temperature of 0.1 eV corresponding to the thermal velocity $v_\mathrm{th} = 2.5$ km s^{-1}, the mean-free path of $2^3\mathrm{S}$ atoms becomes $v_\mathrm{th}/\left(n_\mathrm{e}\langle\sigma v\rangle_\mathrm{eff}\right) =\sim$ 2.5 mm and smaller than the spatial scale of the measurement. A similar analysis is possible also for $2^1\mathrm{S}$ atoms.

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